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Analytics-driven complaint prioritisation via deep learning and multicriteria decision-making

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ABSTRACT

Complaint analysis is an essential business analytics application because complaints have a strong influence on customer satisfaction (CSAT). However, the process of categorising and prioritising complaints manually can be extremely time consuming for large companies. In this paper, we propose a framework for automatic complaint labelling and prioritisation using text analytics and operational research techniques. The labelling step of the training set is performed using a simple weighting approach from the multiple-criteria decision-making (MCDM) literature, while transformer-based deep learning (DL) techniques are used for text classification. We define two priority classes, namely, urgent complaints and other claims, and develop a system for automatic complaint categorisation. Our experimental results show that excellent predictive performance can be achieved with state-of-the-art text classification models. In particular, BETO, a bidirectional encoder representations from transformers (BERT) model trained on a large Spanish corpus, reaches an accuracy (ACCU) and area under the curve (AUC) of 92.1% and 0.9785, respectively. This positive result translates into a successful complaint prioritisation scheme, which improves CSAT and reduces the churn rate.

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1. Introduction

Currently, text analytics and natural language processing (NLP) are trends in business analytics given the pressing need to generate value from the textual information available from companies (Borchert, Coussement, De Caigny, & De Weerdt, 2023; Kriebel & Stitz, 2022). Although deep learning (DL) has not demonstrated improvements in tabular data applications, such as credit scoring (Gunnarsson, Vanden Broucke, Baesens, Óskarsdóttir, & Lemahieu, 2021), there is overwhelming evidence of the virtues of recent DL developments for NLP (Kraus, Feuerriegel, & Oztekin, 2020). These techniques have revolutionised the way companies and other organisations operate because valuable knowledge can be extracted from textual data (Kraus et al., 2020).

Among the various text analytics applications, the automatic analysis of enquiries and claims is a very relevant challenge given its influence on customer satisfaction (CSAT). A good complaint management policy of a company shows customers its ability to listen and react, deal with their concerns efficiently and ensure satisfaction. Companies that are able to manage customer feedback adequately are 6% more profitable and 5% more productive than other companies on average (McAfee, Brynjolfsson, Davenport, Patil, & Barton, 2012). Complaint Analytics (CA) not only enables a swift response to urgent issues regarding service quality, but also an understanding and response to shifts in consumer preferences, the evaluation of CSAT, the delivery of standardised services, and the prevention of customer attrition (Yan, Xu, Tong, & Huang, 2021).

One important challenge in text categorisation is the labelling process. Reading comments and other forms of customer feedback and interpreting them manually are time-consuming processes. In CA, companies create categories for types of requests or customers (Goldman & Gupta, 2018; Wetzel, Hammerschmidt, & Zablah, 2014), as this information can be useful for the automatic labelling process.

This study is motivated by the needs of a Chilean company to define and automate a prioritisation scheme to handle complaints and other claims. Various categories for types of requests and customers can be combined with the adequate input of experts. We propose a weighting model from the multicriteria decision-making

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(MCDM) (Mardani et al., 2015) literature to cast this information into categories to construct a training set. This procedure facilitates complex decision-making, where multiple factors come into play. Once the problem is cast into a binary classification problem with two priority levels (urgent complaints and other claims), we model the text prediction task considering state-of-the-art DL approaches. In particular, we explore the performance of the bidirectional encoder representations from transformers (BERT) model and its variants that are designed to perform the word embedding process during model training (Devlin, Chang, Lee, & Toutanova, 2018).

The proposed framework is presented as a general automated scheme for dealing with claims, enquiries and complaints within any company. This framework was applied to data from a Chilean private nonprofit organisation that provides coverage for occupational accidents and designs risk prevention programs. Chilean law mandates that all employing entities must be affiliated with one of these social security agencies. With more than 73,000 affiliated employing entities and 2.6 million workers, the agency that provides these data is the largest in the country.

The main contributions of this study are as follows:

- There is an important gap between CA and state-of-the-art DL methods. Although there are few machine learning studies devoted to this application (see e.g., Coussement & Van den Poel, 2008), they have two important limitations: (1) the word embedding process is done independently from model training, which leads to a loss of information in contrast to models that "learn" the best embedding based on the data, and (2) they do not consider the possibility of using a pretrained model based on a large source of data, such as Wikipedia. These limitations are important to consider when a relatively small sample of complaints is available to train the model.
- We propose a combination of two different worlds. On the one hand, MCDM is a family of techniques widely used in operational research to address complex decision-making processes. MCDM can be useful as an input for DL because it helps combine business rules to provide a consistent labelling step. In this way, we can summarise multiple factors that define the priority of a complaint as a single score, which is the main methodological contribution of this study.
- Moreover, this study presents a novel complaint management application in which an agency funnels claims made by workers about their employing entities. In contrast, previous work on CA has not involved a third party and managed enquiries directly. Nevertheless, our framework is general enough to be applied in both cases.

The remainder of this paper is structured as follows. Section 2 provides an overview of prior studies on text analytics applied in complaint management. The framework proposed for analysing complaints automatically using DL and MCDM methods is presented in Section 3. Section 4 provides the results obtained with a real-world dataset from a Chilean agency dedicated to providing health services, prevention and coverage of occupational accidents, and economic services. The main conclusions are summarised in Section 5, proposing possible directions for future developments.

2. Prior work on CA

As companies grow in size, the problem of how to respond to a large number of requirements in a timely manner worsens. Medium-sized businesses can receive hundreds or thousands of claims daily, which requires them to have the ability to efficiently address and resolve many complaints. This requirement creates the need for automated CA systems that can prioritise urgent claims over those that can be postponed (Goldman & Gupta, 2018; Singh, Saha, Hasanuzzaman, & Dey, 2021).

Although the most equitable approach to tackling such enquiries is on a first-come, first-serve basis, several studies have suggested that requests should be prioritised and answered based on different guiding criteria. Current practices for complaint prioritisation include the creation of categories for types of requests and/or customers and the setting of service-level agreements to avoid backing up low-priority claims indefinitely. This approach can be carried out by establishing first reply time deadlines or by updating the priority of the claims as time passes (Forrest V. Morgeson, Hult, Mithas, Keiningham, & Fornell, 2020; Wetzel et al., 2014).

The problem with manual complaint prioritisation is that going through each claim and redirecting it to the relevant person or department is a slow process and reduces the time that the marketing team has to focus on the customer experience. Furthermore, complaints are often wrongly classified due to human error, and high-priority complaints can be put aside due to other claims in the queue (Coussement & Van den Poel, 2008; Morgeson et al., 2020).

Several studies addressing CA using text analytics have been proposed in the last two decades. Before DL, *word embedding*, which consists of representing words and documents in such a way that algorithms can make sense of them (Goodfellow, Bengio, Courville, & Bengio, 2016; Vairetti, Martínez, Maldonado, Herrera, & Luzón, 2020), was performed independently from the learning task. For example, the term frequency-inverse document frequency (TF-IDF) approach constructs vectors from single words by analysing how often a word appears in a text in relation to its presence in other texts (Khan, Baharudin, Lee, & Khan, 2010). These vectors represent the input for the learning machine.

In Coussement & Van den Poel (2008), for example, an automatic email classification system was developed to distinguish complaints from claims that are not complaints using TF-IDF. This system combined the unsupervised techniques latent semantic indexing and singular value decomposition for topic mining and dimensionality reduction with the AdaBoost classifier. Similarly, Yang, Xu, Yang, & Chen (2018) considered evidential reasoning rule-based classifiers to analyse customer complaints with uncertain information, while Joung, Jung, Ko, & Kim (2019) used TF-IDF and hierarchical clustering to identify customer needs from complaint data in an unsupervised manner.

With the rise in the use of DL, new architectures have been designed to take advantage of the order in which words are located in a sentence (LeCun, Bengio, & Hinton, 2015), opening new possibilities for CA. Techniques such as recurrent neural networks (RNNs) have been able to incorporate the word embedding process into the network, thus learning the representation of the textual data that best suits the predictive task. RNNs, however, face many challenges in terms of efficiency and performance, which are partially overcome by DL architectures such as long short-term memory (LTSM), gated recurrent units (GRUs), and bidirectional GRUs (BiGRUs) (Malte & Ratadiya, 2019).

The LSTM method was considered in Hake, Rehse, & Fettke (2019), which addressed the problem of complaints on finished products in the world of medical technology manufacturers, where the quality of the final product is rigorously evaluated in the public health environment. Public health issues are always very sensitive due to the significant impact of defective products. In the above research, textual descriptions and sources of errors found from past complaints were treated, which served as input for an automatic suggestion system of possible errors found in future customer complaints. Alternatively, Filgueiras et al. (2019) used LSTM on data from the Portuguese Economic and Food Safety Authority,

with the goal of categorising complaints according to three different classification tasks: target economic activity, implied infraction severity level, and institutional competence.

Currently, the state-of-the-art DL models for text analytics focus on transformers, which are based on the concept of attention (Vaswani et al., 2017). One of the most important transformer-based architectures is BERT, which learns from complete sentences and even paragraphs, word by word, integrating the word embedding in the learning process (Devlin et al., 2018).

The training of the language model is done by masking words randomly. Once approximately 15% of the words are masked in a sentence by the encoder, the decoder aims to predict the missing words. In the same direction, BERT makes sense of pairs of sentences via a next sentence prediction (NSP) approach. The encoder receives a pair of sentences where the second may or may not be subsequent to the first in a paragraph. BERT produces half of a pair of sentences in the right order and another half with two randomly selected sentences from a document. The decoder then aims at predicting whether or not two sentences are in the right order (Devlin et al., 2018).

Another appealing quality of BERT is its ability to learn from a source of textual data different from the task at hand. The whole idea of predicting masked words or NSP is very challenging and requires an enormous amount of data to properly train a model, which can be challenging in tasks such as CA that rely on customer feedback. Fortunately, a "pretrained" model, which includes weights calibrated on large datasets for multipurpose text analysis, can be used. For example, Google made available a pretrained model trained on books and Wikipedia. Next, we can "fine tune" the weights with the data from the downstream task, alleviating the need for a large data source. This process is known as transfer learning (Goodfellow et al., 2016).

Among the recent variants of the BERT model, RoBERTa has gained popularity due to its positive performance. RoBERTa removes the NSP step of BERT and improves masked language modelling to obtain meaning across sentences via masked words only (Liu et al., 2019).

Although there are no studies designed to classify complaints and other claims using transformers, Viellieber & Aßenmacher (2020) considered BERT, RoBERTa, and other variants for automotive complaint analysis. The above authors performed tentative probes for common-sense factual knowledge on pretrained models, adapting these models to a corpus from a specific domain (CA), showing that their pretrained model was able to represent domain-specific knowledge and identifying technical quality issues from the described incidents in customer feedback in the automotive industry.

A different CA approach is the classification of customer reviews into praises, neutral comments, or complaints. Complaints are not always directed at the company, often being in the form of social media comments or reviews. This task is strongly related to sentiment analysis and polarity detection, in contrast to our study and the aforementioned approaches by Coussement & Van den Poel (2008) or Filgueiras et al. (2019), which focus on text classification. In the context of online reviews, the research carried out in Khedkar & Shinde (2020) covered the large amounts of information generated by social networks concerning customer likes and dislikes regarding hotel reviews, mainly using convolutional neural networks (CNNs) and ensembles of traditional machine learning methods. Additionally, other studies focused on classifying opinions in terms of praise and complaints using sentiment analysis in a wide variety of domains and with a wide variety of data sources, including e-commerce product reviews (Yang, Li, Wang, & Sherratt, 2020) and tweets related to different companies (Singh et al., 2021). DL, including CNNs and attention-based BiGRUs, was used in the above studies.

The most relevant CA studies that consider DL are summarised in Table 1. Each publication is classified based on five relevant aspects of this study: (1) whether DL is used, (2) whether BERT is used, (3) whether it is a complaint prioritisation task, (4) whether it uses complaint data instead of other forms of customer feedback, and (5) whether MCDM is used for labelling purposes. The third column presents the best machine learning model(s) for the task at hand.

In summary, the following important aspects that are addressed in the CA literature are key elements in our study:

- Most studies do not analyse complaint data directly and aim to distinguish complaints and other features from other forms of customer feedback data.
- To the best of our knowledge, only one study addresses, although indirectly, the challenge of complaint prioritisation in terms of severity: Filgueiras et al. (2019) predicted infraction severity from complaints in the context of economic and food safety regulations. An infraction can be a simple consumer conflict or a more severe type, such as a crime or administrative infringement. We consider this task to be complaint prioritisation, although it is in a very different domain than ours.
- Few studies consider the current state of the art for DL-based text classification (BERT and other transformer-based models), which is somewhat expected because these strategies were presented only a few years ago, and, for certain industries, it might take a few years to acknowledge their potential. Most methods consider classification strategies that are not state of the art. A second reason is the fact that these methods are considered black boxes, which are not always well considered by decision-makers in the field. In several business applications, such as credit scoring, explainability is of utmost importance (Stevenson, Mues, & Bravo, 2021).
- None of the papers previously discussed consider an MCDM strategy in combination with DL. Furthermore, we are not able to identify any study that uses MCDM for the definition of the labels in the context of text classification. In Kartal, Oztekin, Gunasekaran, & Cebi (2016), the authors proposed an MCDM framework for labelling inventory items in the context of automotive production and warehousing, implementing three well-known MCDM strategies: simple-additive weighting (the strategy considered in our framework), an analytical hierarchy process, and VIekriterijumsko KOmpromisno Rangiranje (VIKOR). The labels were used as input for an inventory classification problem, in which traditional machine learning methods such as artificial neural networks (ANNs) and a support vector machine (SVM) were used.

3. Proposed framework for complaint prioritisation using DL

In this section, we discuss a novel framework for addressing complaint prioritisation via machine learning. Complaints are verbatim comments that can be classified automatically using text analytics tools. In particular, recent advances in DL, such as BERT (Devlin et al., 2018), provide excellent tools for text categorisation, which can be used for the task at hand.

Our main goal is to develop a learning machine that interprets complaints and enquiries, categorising them into different labels that define priorities. These labels can be used as input for complaint routing, assigning them to the proper resolution team according to their level of risk and complexity. This strategy contributes to the analytics literature by providing a novel approach to the automatic classification of complaints, which allows for the identification of improvement gaps in the CSAT. This study also

Table 1
Summary of relevant CA studies.

Publication	Task	Best model(s)	(1)	(2)	(3)	(4)	(5)
Hake et al. (2019)	Defective product detection	CNNs, LSTM	√	✗	✗	√	✗
Filgueiras et al. (2019)	Complaint classification (regulation)	SVM, LSTM	√	✗	√	√	✗
Khedkar & Shinde (2020)	Hotel review classification	CNNs, ensembles	√	✗	✗	✗	✗
Yang et al. (2020)	e-Commerce product review classification	CNNs, GRU	√	✗	✗	✗	✗
Viellieber & Aßenmacher (2020)	Identification of technical quality issues	BERT	√	√	✗	√	✗
Singh et al. (2021)	Tweet classification	BiGRU	√	✗	✗	✗	✗
Our study	Prioritisation of complaints	BERT	√	√	√	√	√

1 Use of DL. 2 Use of BERT. 3 Complaint classification task. 4 Use of complaint data. 5 Use of MCDM for labelling.

contributes to the state of the art of the application of AI because it provides an efficient strategy through which thousands of claims that arrive in real time can be classified. This approach allows for the better use of human resources in organisations, maintaining a focus on customer experience. In concrete terms, an automatic prioritisation system for claims avoids negligence in the assignment of priority made by executives, reducing waiting times and treating each complaint more opportunely according to its seriousness or urgency.

In the simplest version of the framework, we construct a binary classification model that distinguishes between urgent/serious and nonurgent complaints. However, more than two priority levels can be defined, casting the problem into a multiclass one. We suggest the use of an MCDM technique for the labelling process, which is a very useful tool for making complex decisions when multiple criteria exist (Mardani et al., 2015). Before presenting the proposed framework, Section 3.1 describes the MCDM model and its relevant background.

We consider a four-step methodology for analysing complaint data. As a starting point, we consider the cross-industry standard process for data mining (CRISP-DM) model (Chapman et al., 2000), which is a generic framework that has been widely used in machine learning and text analytics studies (Villarroel-Ordenes, Theodoulidis, Burton, Gruber, & Zaki, 2014). Our goal is to adapt the CRISP-DM model, customising it for CA and incorporating novel elements, such as labelling and tokenisation steps.

The first step of the proposed framework consists of data labelling and processing and is presented in Section 3.2. Next, we formalise the complaint prioritisation scheme as a text classification process in Section 3.3. Finally, we discuss how complaint categorisation can be used for decision-making in Section 3.5.

3.1. Preliminaries: MCDM model

The goal of MCDM is to define the importance of each criterion in relation to others, assuming that each decision that is made requires the proper balance of multiple factors (Belton & Stewart, 2002). MCDM methods are capable of involving both quantitative and qualitative factors in a wide range of tasks in fields including energy, environment and sustainability, supply chain management, quality management, geographic information systems (GISs), systems and technology, and tourism (Mardani et al., 2015).

The MCDM process is based on three main components (Mardani et al., 2015):

- Problem structuring: This step consists of understanding the nature of the problem at hand. The interests of the stakeholders related to the problem, possible alternatives, uncertainties, constraints, competition, objectives, and data available are considered inputs.
- Model building: Once the MCDM model is selected and the weightings of the criteria are defined, the model building step consists of the application of the aggregation function. In this step, the best alternative is separated from the other available options. There is a wide variety of MCDM techniques, including simple weighting strategies and more sophisticated strategies. This stage also includes the definition of the criteria used to analyse, evaluate and prioritise the alternatives.
- Challenging thinking: A sensitivity (SENS) and robustness analysis of the solution is carried out, considering the opinion of stakeholders, which allows for the creation of new alternatives, discovery of new challenges, and synthesisation of the relevant insights obtained. This last phase allows for the delivery of value added to companies' business, improving CSAT by identifying gaps in improvement, or simply maximising profits (reducing costs and maximising profits). This last step is not crucial in our framework because we consider the MCDM approach as input for the text analytics model.

Most MCDM applications reported in the literature tend to deal with 5–8 criteria (Mardani et al., 2015).

Arguably, the best-known MCDM model is the weighted-sum model (WSM) (Mardani et al., 2015), which we propose for the definition of the priorities of each complaint. However, we discuss more sophisticated variants in case the prioritisation process requires a more complex scheme.

Given m alternatives and n decision criteria, the WSM score is defined as follows:

$$A_i^{WSMscore} = \sum_{j=1}^{n} w_j * a_{i,j} \quad for \quad i = 1, \ldots, m, \tag{1}$$

where $A_i^{WSMscore}$ represents the score associated with the weighted sum of the value of alternative $i = 1, \ldots, m$ on criteria $j = 1, \ldots, n$, with a weight, w_j, associated with criterion j.

The underlying idea is simple: each alternative is evaluated according to multiple criteria, which have a bearing on the evaluation of the alternative. This method is easy and fast to implement and very transparent for decision-makers (Mardani et al., 2015).

The assignment of the weights for each criterion is carried out according to the strategies established *a priori*, which can be established from a stakeholder depending on the nature of the problem. Alternatively, a group of different actors involved in the problem can define the weights through a consensus or even a different approach altogether.

Alternatively, there are more sophisticated methods for the computation of the score associated with each alternative, which allows us to capture more complex underlying patterns associated with the problem. Among the most popular strategies are the analytic hierarchy process (AHP), technique for order of preference by similarity (TOPSIS), Elimination et Choix Traduisant la Realité (ELECTRE), hybrid MCDM, analytic network process (ANP), preference ranking organisation method for enrichment evaluation (PROMETHEE), and VIKOR (Mardani et al., 2015).

3.2. Complaint intake, labelling, and processing

The first step consists of providing a suitable environment for complaint/enquiry intake, which is the process of gathering initial information from the claim and complainant. A centralised intake system is important for the consolidation of the required information across communication channels.

Since intake is a crucial step in complaint analysis, it should be adequately staffed. Some companies have one or more agents dedicated solely to intake, while other companies rotate intake tasks among executives. Furthermore, intake agents may attempt to resolve complaints/enquiries before initiating the formal intake process (Goldman & Gupta, 2018). Since our goal is to automate the creation of variables and the triage process by using text analytics techniques directly on the verbatim text, important savings can be achieved in terms of executive time.

The intake step involves the following stages: (1) assessment of data quality via descriptive statistics, identifying problems with textual data, (2) finding of the first patterns in the data that can be useful for decision-making in CA, (3) preparation of the textual data for the complaint classification task, and (4) construction of labels for complaint prioritisation.

Regarding the third point, traditional NLP techniques such as stemming (removing the last few characters from a word), lemmatisation (converting a word into its meaningful base form), and tokenisation, which is a key step in our framework, in which a piece of sensitive information is replaced by a surrogate value for security purposes. It is important to note that we go beyond traditional tokenisation and propose the designing of generic tokens for different pieces of information (ID numbers, customer IDs, dates, times of day, emails, and telephone numbers) to be recognised by the language model. The purpose of this approach is to improve predictive performance rather than to mask sensitive information.

Regarding the fourth point, a key contribution of the proposed framework is the definitions of the labels (urgent or nonurgent complaints or claims). Agents within the company can determine the severity of a claim based on a series of factors, which allows for a proper definition of the weights of the MCDM model described in Section 3.1. These factors essentially constitute manually-generated metadata derived from the input text. The main purpose of the machine learning model is to automate the priorisation scheme by learning an aggregation of these factors via supervised learning. In doing so, an organization can bypass the potentially time-consuming tasks associated with metadata creation and the subsequent triage process. It is important to note that these factors are not utilized as inputs in our framework but are employed solely for label generation. Once the weights of the MCDM model are defined, one or several thresholds can be set and used to bin the MCDM score. One threshold results in a binary classification model, while more thresholds can be considered for the definition of an ordinal classification problem with various priority levels.

3.3. Learning process

Once the intake process is completed, a claim should be triaged. In this study, the task of classifying the urgency of complaints and enquiries is considered, in which they are labelled using the MCDM strategy. We suggest the use of state-of-the-art DL techniques for model training and the subsequent prediction of new complaints and enquiries, the objective of which is to determine whether or not a given claim should be prioritised to treat it in a timely manner according to its degree of importance.

We propose the use of BERT and other variants for the word embedding and learning processes. We can explore different pretrained models depending on the language of the claims (our application consists of complaints and enquiries in Spanish) and then fine tune the model with the data collected for the task at hand.

Regarding running times, note that both the masked language modelling and NSP steps are required only for the training of the pretrained model and are not necessary for the training of the downstream task. The training of the language model is extremely time consuming, but fortunately, it is carried out by the providers of the pretrained models. The text classification task (fine-tuning step) is less time consuming; however, the fine tuning of a medium-sized model (approximately 45,000 samples) can take several hours but can be done offline and as fast as other traditional machine learning techniques. Finally, the application of the trained model can be performed in real time because it consists of the evaluation of a function on new samples (complaints or other claims). In summary, implementing large language models (LLMs) is computationally expensive for businesses but similar to other machine learning methods. The text classification task is performed only once, leading to a model that can be applied, even in real time, to new complaints and enquiries.

For the binary classification task of urgent/nonurgent complaints and enquiries, we recommend sigmoid activation in the output and binary cross-entropy as the loss function for the last layer of the model. Alternatively, the complaint prioritisation task can be a multiclass problem with different levels of urgency. In this case, a SoftMax activation function with (multiclass) cross-entropy loss is recommended. Finally, this task can also be seen as a regression problem with a numerical output. For example, urgency can be defined on a 5- or 10-point scale. Here, the mean squared error (MSE) can be used as the loss function (Goodfellow et al., 2016).

3.4. Model evaluation and implementation

Regarding model validation, the fine-tuning step requires a process to define the optimal hyperparameter configuration. We consider a standard procedure in which we divide the dataset into training, validation, and test subsets using stratified random sampling. In this way, we guarantee that the two labels (urgent and nonurgent claims and complaints) are represented in the same proportion in each subset. We first split the data by selecting 90% of the dataset for training and the remaining 10% for testing. The latter set remains unseen during the learning process. Next, 70% of the training set is used for the construction of the model, while the remaining 30% is used for model selection and validation.

For model evaluation, we propose the computing of the accuracy (ACCU), sensitivity (SENS), specificity (SPEC), and area under the curve (AUC) metrics. These measures are widely used in binary classification (HaCohen-Kerner, Dilmon, Hone, & Ben-Basan, 2019). ACCU corresponds to the fraction of correctly classified claims, while SENS and SPEC correspond to the proportion of urgent and nonurgent complaints correctly identified. Finally, the AUC measure is computed as the area under the receiver operating characteristic (ROC) curve.

Regarding the implementation of the triage system, some important points must be taken into account. Several companies receive more complaints and claims than they have the resources to process. Every claim is consequential to the customer, but the urgency of each allegation can differ significantly across complainants and claim types (Goldman & Gupta, 2018).

An automated triage system allows companies to sort claims consistently into different categories to efficiently manage them, but to do so, it requires a new process to deal with complaints and enquiries properly. The functions of the team of agents that handle complaints and enquiries must be redefined, assigning the most experienced executives to urgent complaints and enquiries. Furthermore, customer service metrics and levels of service must

be defined for the various priority categories. Some relevant metrics for CA are average response time, first contact resolution rate, and next issue avoidance (Goldman & Gupta, 2018; Morgeson et al., 2020; Wetzel et al., 2014).

One drawback of an automated triage system is the monetary cost of the analytics project, which can be an important fraction of the budget for small to medium-sized companies. Furthermore, such a system requires some adjustments in the current complaint intake and resolution process, as indicated in the previous paragraph. However, this system can be strongly beneficial for companies that handle a large number of claims per month. The opportunity cost of the analysts and agents that perform manual intake should be considered when estimating the benefits of the proposed framework.

3.5. Complaint resolution and decision-making

In this step, we discuss the decision process behind the model and the opportunities that provide the complaint prioritisation model for decision-making.

Urgent/nonurgent claim classification allows for the adequate routing of complaints and enquiries to the corresponding resolution teams. The automatic risk and complexity assessment of a claim based on its content helps it be addressed in a timely matter.

In a contact centre setting, for example, low-risk claims can be addressed by inexperienced agents, who can follow a standard script for conflict resolution. In contrast, complex and urgent complaints can be directed to skilled agents automatically, which is a key aspect of the model.

Once the complaint is routed to the appropriate executive, the resolution is communicated to the customer as applicable. Reports can be created based on the data collected from the postresolution process to improve complaint management (Davidow, 2003).

Note that process automation is a task in and of itself, which can be extremely useful for saving analysts time. The following other relevant decisions within a company for which our model can be useful exist:

- Churn prediction and loyalty: Noncomplainers, compared to dissatisfied customers, have a low chance of leaving the company (Blodgett & Anderson, 2000). In contrast, a satisfied complainant will strengthen his or her ties with the company and therefore his or her loyalty, which translates into positive word of mouth (Umashankar, Ward, & Dahl, 2017). The number of complaints and their severity, together with satisfaction data from the postresolution process, are a very valuable source of information that can act as input for a churn prediction model. Such a model allows for the design of customised retention campaigns, which can be linked to the nature of the complaints made by customers (Yan et al., 2021).
- Profiling: The differences between customers in terms of how they complain are an interesting input for creating profiles (Joung et al., 2019). Complaint management data can complement other traditional data sources, such as demographics or previous purchase behaviour, in constructing customer portraits. Profiling allows for the design of tailored products and services to specific groups of customers; however, attention can also be customised in the sense that premium consumers can be treated differently in terms of complaint resolution.
- Staffing: Understanding the complexity and urgency of complaints can improve staffing decisions. In a contact centre, for example, we want not only to estimate the demand for both urgent and nonurgent complaints accurately but also to manage the rotation of agents to meet this demand with more suitable resources.

Finally, Fig. 1 summarises the proposed framework in the form of a flowchart. The sequence of the step is not strict, allowing revisiting previous steps. Arrows highlight key dependencies between steps, aligning with the CRISP-DM framework (Chapman et al., 2000).

4. Results and managerial insights

In this section, the company that provides the data is first presented in Section 4.1, explaining the procedure followed for the prioritisation scheme. Next, Section 4.2 reports a summary of the main results of the classification framework proposed for CA.

4.1. Data and experimental setting

In Chile, one of the major problems that has caused the industrialisation and automation of productive processes is occupational accidents. For this reason, law no. 16,744 was enacted in 1968, which established the personal obligation of employees to have insurance for occupational accidents and diseases. This insurance commits employers to bear the health costs of their workers. Within this regulatory framework, security mutuals, of which the largest is the Chilean Safety Association (ACHS), with more than 2,600,000 affiliated workers, were created.

The government of Chile, through the Superintendency of Social Security (or SUSESO), is in charge of evaluating and supervising the care received by security mutual affiliates. This audit considers important requirements that must be met by the firm in terms of service quality.

Due to the growth of the Chilean population and its workforce, having a good relationship with their affiliates and meeting their expectations has become a challenging task for organisations. Therefore, an automated complaint management system is of utmost utility to provide prompt responses to urgent claims, meeting the requirements of the regulator. Given its role as a nonprofit institution that oversees and improves the relationship between workers and employers, a security mutual acts as a third party, or mediator, between workers who file complaints against their employers (e.g., due to poor working conditions causing occupational accidents) and the attention they receive in the medical centre. The security mutual also channels enquiries and other claims.

We use a dataset that encompasses 44,250 complaints and other enquiries collected from the ACHS in 2019. This information is manually categorised into several variables and includes the verbatim text of the claims. From the set of variables, eight are selected to define the prioritisation based on MCDM. This selection is made according to a discussion with the various managers within the company, some of which were considered more important than others in defining the urgency of a claim, making the task suitable for a weighted MCDM scheme.

Each level of the categorical variables used for the prioritisation scheme receives a score assigned by the experts. This score, ranging from 0 to 100, indicates the urgency of the claim for a given factor. The following categorical variables are considered, with their corresponding weights:

- Nature of the claim or request ($w_1 = 0.37$): A claim can be labelled either a complaint (score=100) or an enquiry (score=0).
- Service provision ($w_2 = 0.18$): The enquirer-requested service regarding the security mutual is classified as medical (score=80–100), economic (score=40–60), or other (score=0–20). For example, emergency care receives a score

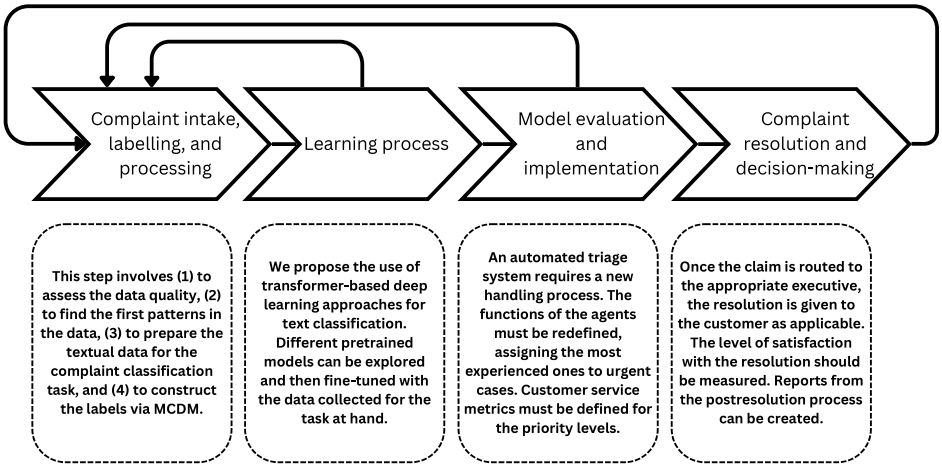


Fig. 1. Flowchart of the proposed four-step framework for analysing and prioritising complaints and enquiries.

of 100, outpatient care receives a score of 80, and the calculation of severance payments receives a score of 60.

- Service priority ($w_3 = 0.12$): The priority of the service provided is classified into the first priority (score=80–100, mainly hospital care services), second priority (score=40–60, for example, rehabilitation therapy and industrial hygiene), and third priority (score=0–20, for example, certificates and other administrative requests).
- On-site claim ($w_4 = 0.09$): The enquiry can be made either on site (score=100) or off site (score=0).
- Employer segment ($w_5 = 0.07$): The security mutual defines different employer segments, where large customers have a score of 100, independent employees and small firms have a score of 50, and others have a score of 0.
- Communication channel ($w_6 = 0.06$): This is a second classification for employers and distinguishes among communication channels from large customers (score=100), medical centres (score=80), or other channels (score=40).
- Nature of the enquirer ($w_7 = 0.05$): The enquirer can be an affiliated with a security mutual (score=100), a retired person (score=70), or someone not affiliated with the security mutual (score=0).
- Nature of the cause ($w_8 = 0.05$): Similar to the previous category, the party that causes the service provided can be someone affiliated with the security mutual (score=100), a retired person (score=70), or someone unaffiliated with the security mutual (score=0).

Based on this information, the WSM score is computed for each claim. Finally, a threshold is defined to determine whether or not a claim is urgent. This threshold is set to 50, meaning that a WSM score larger than 50 is considered a high-priority request. The resulting indicator variable is used as the label vector for the text classification task. Since the first factor (nature of the claim or request) is of utmost importance for the company, all the high-priority requests correspond to complaints.

It is crucial to note that these seven factors are not treated as covariates in the input matrix, but are used exclusively to construct the target variable. The textual data are the sole inputs. Thus, no data leakage occurs during the testing process.

Changing the weights of the MCDM model results in different datasets, affecting the performance of the various machine learning classifiers. We can even optimise these weights to improve ACCU. However, this approach deviates from the original goal: the optimal balance between factors should be given by experts to define the priority of the complaints and claims. In other words, the weights of the MCDM are not hyperparameters that must be tuned.

To implement the preprocessing strategies (stop word removal, stemming, and TF-IDF embedding) and machine learning methods, Python is used. The nltk, sklearn, and keras libraries are considered for NLP, machine learning, and DL modelling, respectively. DL strategies are also implemented using the Flax and JAX platforms.

4.2. Result summary

The plots in Figs. 2 and 3 illustrate the distribution of the three most relevant variables for the MCDM framework. Since complaints account for approximately half of the requests processed by the company, the proportions in terms of service provision are relatively similar for the three levels of this variable (medical, economic, or other) and for the two types of requests (complaints and enquiries; see Fig. 2). In contrast, the overwhelming majority of requests are from the first service priority (hospital care services, see Fig. 3).

The MCDM indicator is computed according to Eq. (1) and then discretised into a binary variable. A total of 21,015 of the sample complaints or claims are considered urgent (47.49%), while the remaining 23,235 are considered nonurgent (52.51%).

After analysing the various claims and requests, we note that Chilean ID numbers, customer IDs, dates, time of the day, emails, and telephone numbers appear in a large proportion. This information can confuse NLP methods by introducing noise into the models. Thus, we consider a tokenisation step for this information, using a single token for all Chilean ID numbers, another token for customer IDs, and so on for the six previously mentioned sources of noise. In this way, the classification model can recognise these patterns easily. To avoid a potentially time-consuming process of manual replacement, Python's regex library is used for data imputation. For consistency, the tokenisation step is applied to the whole dataset.

The results for the various text classification approaches considered in this study are as follows:

- Logistic regression + TF-IDF (Logit): We consider several traditional classifiers along with the TD-IDF discussed in Section 2, which constructs the word embeddings as a preprocessing step. Although this strategy is not state of the art, we present the results for completeness. Among the various traditional classifiers, logistic regression is a statistical method that produces positive predictive results without the need for hyperparameter tuning. This technique computes a

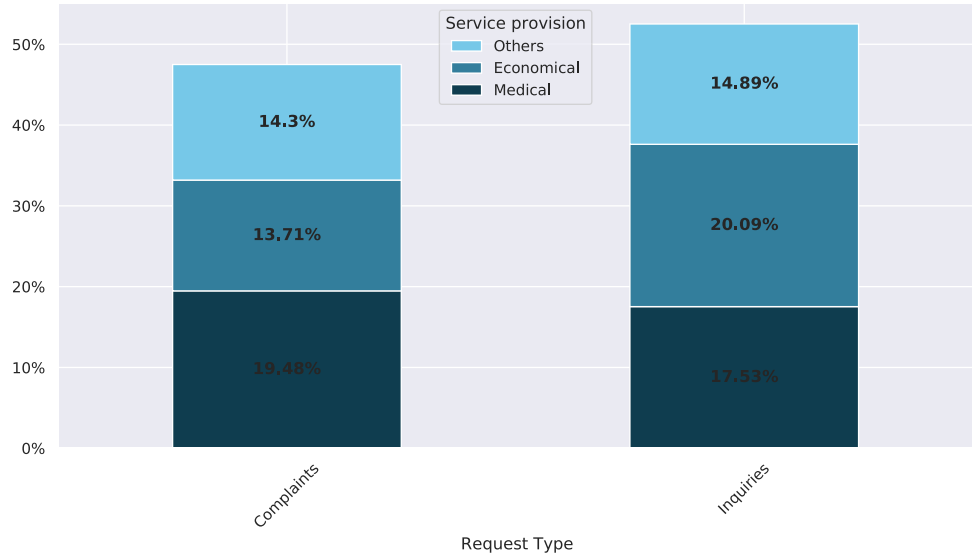


Fig. 2. Nature of the claim or request (complaint or enquiry) disaggregated by service provision (medical, economic, or other).

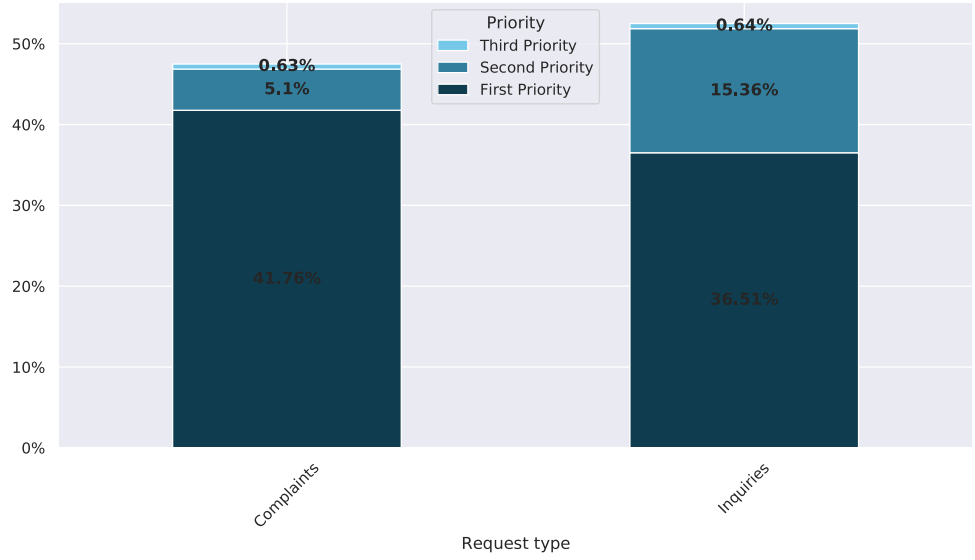


Fig. 3. Nature of the claim or request (complaint or nquiry) disaggregated by service priority (first, second, or third). Notice that service priority is one of the seven factors or metadata created by the analysts that represent the input of the MCDM model, and does not represent the target variable (urgent/nonurgent queries).

weighted sum of the covariates and vector coefficients $\boldsymbol{\beta}$, resulting in a separating hyperplane.

- Naïve Bayes + TF-IDF (N. Bayes): Another statistical method for classification is Naïve Bayes, which estimates the posterior probability of a given class using Bayes theorem. This method assumes class-conditional independence; i.e., the effect of the value of a variable on the class is independent of the values of the other variables.
- Multilayer perceptron (MLP) + TF-IDF (MLP): We also consider the traditional (shallow) neural network architecture using a single hidden layer. This approach can construct nonlinear decision surfaces, leading to potentially better results for other traditional classifiers. However, this approach has a higher risk of overfitting.
- SVM + TF-IDF (SVM): This statistical learning method balances the structural risk and model fit by considering l_2 regularisation and the hinge loss function. This technique can construct nonlinear decision boundaries due to the use of kernel functions, which have a lower risk of overfitting compared to MLP.
- BERT-base-multilingual-cased (BERT(b)) (Devlin et al., 2018): We explore five different state-of-the-art DL techniques for text classification in Spanish. The first is BERT multilingual, which was pretrained on 104 languages. This model is case sensitive. The BERT base(b) model has 12 encoder layers stacked on top of each other, in contrast to BERT large(l), which has 24 layers of encoders.
- BETO-base-cased (BETO(b)) (Cañete et al., 2020): This model corresponds to the BERT base trained on a large Spanish corpus developed in Chile, which is also case sensitive.
- RoBERTa-base-xlm (RoBERTa(b)) (Liu et al., 2019): This model is the multilingual RoBERTa model, which considers 12 encoder layers.
- BERTIN-RoBERTa-base (BERTIN(b)) (la Rosa et al., 2022): BERTIN is a series of BERT and RoBERTa variants in Spanish and developed in Spain. This strategy considers the RoBERTa base model.
- BERTIN-RoBERTa-large (BERTIN(l)) (la Rosa et al., 2022): In contrast to BERTIN(b), this method considers the large RoBERTa model with 24 encoder layers.

Table 2
Performance summary of the various text classification models. ACCU, SENS, SPEC, and AUC metrics are reported. The AUC measure is multiplied by 100.

Classifier	ACCU	SENS	SPEC	AUCx100
BETO(b)	**92.10**	92.26	**91.85**	**97.85**
RoBERTa(b)	92.00	**93.20**	90.57	97.62
BERTIN(b)	91.30	92.21	90.23	97.43
BERT(b)	91.50	92.04	90.96	97.43
BERTIN(l)	90.10	91.05	89.10	96.90
SVM	90.21	90.11	88.51	96.38
Logit	89.79	89.66	87.97	96.59
MLP	89.04	89.41	87.82	95.36
N. Bayes	86.65	88.04	86.54	94.70

Table 3
Performance summary of the various text classification models without tokenisation. ACCU, SENS, SPEC, and AUC metrics are reported. The AUC measure is multiplied by 100.

Classifier	ACCU	SENS	SPEC	AUCx100
BETO(b)	**91.10**	91.23	**90.95**	**97.53**
RoBERTa(b)	91.00	**91.78**	90.15	97.38
BERTIN(b)	90.50	91.11	89.75	96.99
BERT(b)	90.40	90.89	89.80	97.10
BERTIN(l)	89.30	90.17	88.19	97.42
SVM	88.53	89.35	87.39	95.36
Logit	87.84	88.00	85.54	95.78
MLP	87.18	88.60	86.69	94.15
N. Bayes	85.21	87.24	85.29	93.94

Regarding the hyperparameters, all DL models are trained using a batch size of 32, a weight decay of $\{0.01, 0.1\}$, and up to 5 epochs. The AdamW optimiser is considered with $\beta_1 = 0.99$, $\beta_2 = 0.999$, and $\epsilon = 0.000001$. For the alternative methods, linear SVM and logistic classifiers are constructed with $C = 1$ and l_2 regularisation, and the number of hidden units and epochs are set to 100 and 300, respectively, for the MLP model.

Table 2 reports the results of the experimental analysis, showing the ACCU, SENS, SPEC, and AUC metrics for the various classification methods. We consider the AUC the main performance metric because it does not depend on the final threshold: the curve defines the relation between SENS and the rate of false alarms (1-SPEC) for all possible thresholds that can be obtained from a probabilistic output (Goodfellow et al., 2016). Note that the use of sigmoid activation leads to an output that can be interpreted as the probability of a claim being urgent. The best result (largest value) is highlighted in bold for each metric.

Table 3 shows that all models are close to 100 in the four metrics, being successful in automatically identifying the most sensitive claims according to the prioritisation scheme. Understanding and automatically classifying complaints in terms of urgency is the main goal of this study, and our results confirm that it is possible to achieve excellent performance with machine learning methods.

As expected, the best classifiers are the five state-of-the-art DL techniques tailored for text classification. However, the difference between them and the traditional machine learning techniques considered in combination with the TF-IDF framework is small. All methods are able to achieve a good balance in terms of SENS and SPEC.

According to Table 2, BETO is the model with the best performance in terms of both ACCU (92.1%) and AUC (97.85). The reason for this may be that BETO was developed in Chile and may have been pretrained with more text written in Chile than may have the other BERT variants. The second-best model in terms of ACCU is the multilingual RoBERTa (base), while BERTIN (large) is the second-best model in terms of the AUC. All the BERT variants achieve a relatively similar performance in terms of both metrics.

To understand the important terms that define the two priority levels, we implement two feature ranking techniques based on the TF-IDF approach. We sort the variables according to 1) the chi-square statistic and 2) the weights of the logistic regression. The top 10 terms in terms of importance are as follows:

- Chi-square statistic: 1) company, 2) certificate, 3) folio, 4) report, 5) worker, 6) subsidy, 7) rate, 8) request, 9) account, and 10) income.
- Logistic regression: 1) complaint, 2) disturb, 3) agreement, 4) qualification, 5) disagreement, 6) appeal, 7) care, 8) pain, 9) negative, and 10) verify.

Interestingly, the rank provided by the chi-square statistic considers terms that are related mostly with formalities, i.e., nonurgent claims. Concepts such as “certificate”, “folio”, “report”, “rate”, or “request” are included in administrative queries related to nonmedical service provisions. In contrast, the rank established by the logit model includes terms associated with (urgent) medical complaints, including “disturb”, “care”, or “pain”. As expected, the word “complaint” is relevant because enquirers tend to use it in their reports.

For comparison purposes, the performance of all models without the tokenisation step is presented in Table 3. We consider the same performance metrics (ACCU, SENS, SPEC, and the AUC) for the various classifiers. The best result is highlighted in bold for each performance measure.

According to Table 3, the performance of all models is improved with the tokenisation step, showing the virtues of the approach. We recommend this strategy in case the textual information includes personal information that can introduce noise to the models. This approach goes beyond the usual tokenisation step to anonymise sensitive information for privacy issues. Furthermore, both approaches can be applied without problems.

5. Discussion and conclusions

In this study, we consider a novel text-mining framework to categorise enquiries and complaints automatically in terms of urgency. Our goal is twofold: to extract knowledge from textual information to improve decision-making and to show that it is possible to automate a potentially time-consuming labelling process of complaints while incorporating expert knowledge. This knowledge is modelled via MCDM methods, which is a very useful operational research tool that allows for the aggregation of relevant criteria for decision-making.

The identification of urgent complaints and defining of a prioritisation scheme to provide prompt responses are key to enhancing the customer experience in large companies. First, loyalty can be enhanced, and customer attrition can be reduced, thus directly impacting profitability. Engaged customers are known to generate approximately 1.7 times more revenue than are other types of customers, while an increase in the retention rate of 5% can reduce operational expenses by up to 18% (Fleming & Asplund, 2007; Maldonado, López, & Vairetti, 2020). Furthermore, understanding complaint data can be a valuable source of information for customer profiling (Joung et al., 2019), which enhances service customisation.

For the particular case of the company that provided the data, complaints are collected from its contact centre to improve staffing decisions. Experienced agents are more suitable than are new agents for dealing with urgent claims. The proposed complaint management framework is designed to be a useful tool for agent scheduling and to channel claims to the right agent.

Experiments on a valuable dataset from a security mutual agency in Chile are carried out. We use contact centre information that include the verbatim quotations of enquiries and complaints, together with variables created manually by the company, for categorisation purposes. The latter is used for the MCDM framework,

which consists of a WSM with a set of eight nominal variables. For each variable and level, a score that defines the urgency is assigned by managers within the company. The relative importance of the variables is also estimated, which allows us to obtain the model weights.

As a rough estimate, the security mutual handles approximately 3300 enquiries and complaints per month on average. Assuming that an analyst spends approximately five minutes creating the seven variables that define the priority of the claim, this is a gain of approximately 275 hours of analyst time. In Chile, a full-time employee works 176 hours per month. Therefore, the savings in terms of analyst time correspond to approximately 1.5 full-time executives.

The MCDM framework provides the labels for a text classification model used to infer the urgency of a complaint. We consider traditional and state-of-the-art DL classifiers for NLP, resulting in a comprehensible empirical study on business analytics based on textual data. This study expands other research on CA based on traditional machine learning techniques (see, e.g., Khan et al., 2010; Kumar & Zymbler, 2019) and DL (see, e.g., Ando, Masumura, Kamiyama, Kobashikawa, & Aono, 2017; Ramaswamy & DeClerck, 2018) by incorporating recent attention-based models, such as RoBERTa. Furthermore, this is the first study to consider MCDM techniques for complaint categorisation.

The main conclusion drawn from our experiments is that attention-based methods are successful in classifying complaints, achieving a better predictive performance than that of traditional machine learning methods. We illustrate the importance of a suitable pretrained method to alleviate the need for textual data in training. In particular, BETO, a BERT model trained on a large Spanish corpus developed in Chile, achieves slightly better performance than do other BERT and RoBERTa variants.

A second important conclusion is that data preprocessing is an extremely important step in text classification. We show that a tailored tokenisation step designed to reduce noise from ID numbers, dates, emails, and telephone numbers can lead to important gains in ACCU. In particular, we improve the ACCU and AUC of the models by one or two percentage points, which is larger than the difference in performance for the four best classifiers. This result indicates that feature engineering and preprocessing can be more effective than can an exhaustive evaluation of different state-of-the-art methods.

The proposed text classification tool is designed to automate a manual labelling process of claims made by agents, including a prioritisation scheme defined by business experts. Experiments are performed on a curated dataset based on manual transcriptions of enquiries and complaints made by phone through the contact centre. The level of automation can be increased by considering written complaints collected via bots or web forms or using automatic speech recognition (ASR) methods when dealing with oral claims. Although these approaches face more data issues, such as grammar and orthographic errors in written enquiries or transcription mistakes in the case of ASR, we believe that it is possible to accurately prioritise enquiries from these data sources. We consider such a study an interesting alternative for future research.

There are some limitations to this study that can be addressed in future research. The goal of this study is to show that the potentially time-consuming processes of manual categorisation and prioritisation of complaints and other claims can be automated via DL and NLP. Although this goal is successfully achieved with our experiments, we are unable to perform a thorough comparison between former practices and the new system and to quantify the impact of the proposed framework. Such a comparison requires information about salaries, opportunity costs of the team, and other data that we are not able to obtain from the company. Furthermore, we indicate in Section 3.5 how the framework can improve decision-making in the company, achieving additional gains and going beyond automation through, for example, churn management, profiling and staffing decisions. Unfortunately, we cannot assess the impact of the framework empirically for these applications because the necessary data are not available. Given the nature of security mutuals as nonprofit private institutions that are mandatory for workers and employers, they are not as relevant as in other business domains. We consider such analyses as potential future research opportunities.

In the experimental framework, we emphasise on distinguishing complaints and enquires mainly because the company that provided the data requested this approach. In a contact centre, it can be relatively easy to triage complaints and enquires by defining an option in the voice menu or asking the operator to include this information. Alternatively, our framework can be used with complaint data exclusively, establishing two or more priority levels according to the other factors considered in the MCDM function. Notice that all these changes are possible without modifying the proposed framework. We require only a different definition of the weights and the use of more than one threshold for a multiclass problem. The input dataset changes because only the subset of the complaints is now considered. Therefore, the model needs to be retrained. We believe this is an interesting avenue for future research.

Regarding future developments, further opportunities can be identified from this study. First, the text classification approach can be extended easily to multiclass classification or ordinal regression when different urgency levels are considered or when the target variable is treated as ordinal or numerical. While multiclass classification can be useful for identifying complaints in customer reviews via sentiment analysis (positive, neutral, and negative reviews), ordinal/numerical labels can be used for more complex attitudinal characters from decision-makers (high, medium, or low priority, for example). This approach can be accompanied by more sophisticated MCDM approaches. Finally, the drivers of customer experience can be identified in the enquiries and complaints via multilabel classification, as proposed in Aldunate, Maldonado, Vairetti, & Armelini (2022) for customer feedback data.

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